WORKED SOLUTION

STEP II 2007, Question 9

Mechanics • Oblique Impact with Restitution

STEP II 2007, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q9

The key idea

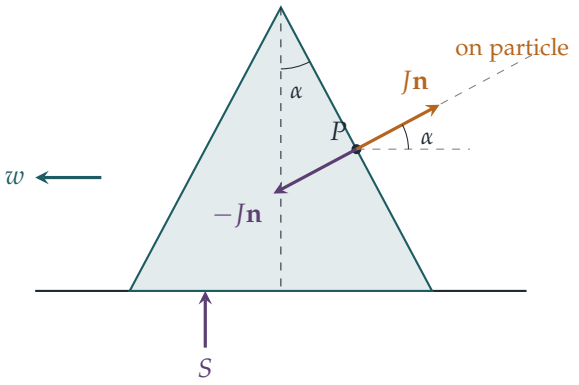
Everything is smooth, so the only impulse between the particle and the cone acts along the common normal at the point of impact, and the table can only push vertically. That tells us two things at once: the particle's velocity *along* the surface is unchanged, and horizontal momentum of the particle and cone together is conserved. Combine that with Newton's law of restitution along the normal and the cone's speed drops out. Part (ii) is part (i) applied to the normal component of the falling particle's velocity.

Solution

Model the particle as a point and the cone as a rigid body that (as we are told) neither rotates nor moves vertically, so just after impact it slides horizontally with some speed w. Work in the vertical plane through the axis of the cone and the point of impact P. Since the generator through P makes angle α with the vertical axis, the outward normal at P makes angle α with the *horizontal*. Take unit vectors

$$\mathbf{n} = (\cos\alpha,\ \sin\alpha) \text{ (outward normal)}, \qquad \mathbf{t} = (\sin\alpha,\ -\cos\alpha) \text{ (down the slope)},$$

with the first coordinate horizontal (away from the axis) and the second vertically upwards.



Let the impulse on the particle be $J\mathbf{n}$ ($J > 0$), so the cone receives $-J\mathbf{n}$, whose horizontal part is $J\cos\alpha$, directed away from P. The table supplies a vertical impulse $S = J\sin\alpha$ upwards, which is exactly what is needed to cancel the downward part of $-J\mathbf{n}$; that is why the cone has no vertical velocity. So

$$Mw = J\cos\alpha, \tag{1}$$

with the cone moving horizontally away from P.

Part (i)

The particle arrives with velocity $-u\mathbf{n}$, which has no component along $\mathbf{t}$. The surfaces are smooth, so the impulse on the particle is purely along $\mathbf{n}$ and cannot change its component along $\mathbf{t}$. That component is therefore still zero after the impact, and the particle's velocity is a multiple of $\mathbf{n}$: it moves along the normal line at P (outwards, or still inwards but more slowly if the cone runs away from it fast enough).

After impact the particle's velocity is $(-u + J/m)\,\mathbf{n}$. The cone moves with speed w horizontally away from P (to the left in the diagram), so its velocity is $(-w, 0)$, whose component along $\mathbf{n}$ is $-w\cos\alpha$.

> **Ask yourself** Newton's law of restitution compares relative velocities along which direction, and what is the cone's contribution?

Restitution acts along the common normal $\mathbf{n}$. Before impact the speed of approach along $\mathbf{n}$ is u (the cone is at rest). After impact the speed of separation is the particle's $\mathbf{n}$-component minus the cone's:

$$\left(-u + \frac{J}{m}\right) - (-w\cos\alpha) = eu \quad\Longrightarrow\quad J = m\big((1+e)u - w\cos\alpha\big).$$

Substituting into (1),

$$Mw = m\cos\alpha\,\big((1+e)u - w\cos\alpha\big) \quad\Longrightarrow\quad \boxed{w = \frac{m(1+e)u\cos\alpha}{M + m\cos^2\alpha}}.$$

Equivalently, horizontal momentum is conserved (the table's impulse is vertical): $m(-u\cos\alpha) = m\big(-u + \frac{J}{m}\big)\cos\alpha - Mw$, which is (1) again.

Part (ii)

Now the particle arrives with velocity $(0, -u)$. Resolve it:

$$(0,-u)\cdot\mathbf{n} = -u\sin\alpha, \qquad (0,-u)\cdot\mathbf{t} = u\cos\alpha.$$

The tangential part $u\cos\alpha$ is unaffected by the impact (smooth surfaces again) and plays no further role: the impulse $J\mathbf{n}$, equation (1) and the restitution equation involve only normal components. So the working of part (i) applies unchanged with the normal approach speed u replaced by $u\sin\alpha$:

$$w = \frac{m(1+e)(u\sin\alpha)\cos\alpha}{M+m\cos^2\alpha} = \frac{mu(1+e)\sin\alpha\cos\alpha}{M+m\cos^2\alpha},$$

as required.

Maximising over α. With M, m, e, u fixed we need to maximise $f(\alpha) = \dfrac{\sin\alpha\cos\alpha}{M+m\cos^2\alpha}$ for $0 < \alpha < \frac{\pi}{2}$.

Ask yourself Differentiating this quotient is messy. Is there a substitution that makes it tidy?

Divide the top and bottom by $\cos^2\alpha$ and put $T = \tan\alpha > 0$, using $\sec^2\alpha = 1 + T^2$:

$$f = \frac{\tan\alpha}{M\sec^2\alpha + m} = \frac{T}{MT^2 + (M+m)} = \frac{1}{MT + \dfrac{M+m}{T}}.$$

By the AM–GM inequality (both terms are positive),

$$MT + \frac{M+m}{T} \geq 2\sqrt{M(M+m)},$$

with equality exactly when $MT = (M+m)/T$, i.e. $T^2 = (M+m)/M$. So f, and hence w, is greatest there. Since $T \mapsto \alpha$ is a bijection from $(0,\infty)$ to $(0, \frac{\pi}{2})$ this is the unique maximum, and

$$\cos^2\alpha = \frac{1}{1+\tan^2\alpha} = \frac{1}{1+\frac{M+m}{M}} = \frac{M}{2M+m}, \qquad \text{so} \qquad \cos\alpha = \sqrt{\frac{M}{2M+m}}$$

(taking the positive root as α is acute). The greatest speed is $w_{\max} = \dfrac{mu(1+e)}{2\sqrt{M(M+m)}}$.

Checks

- If $\alpha \to \frac{\pi}{2}$ the cone is a flat disc: a vertical fall gives no horizontal push and $w \to 0$. In part (i) with $\alpha \to \frac{\pi}{2}$ the normal is vertical, and again $w \to 0$. Good.
- In part (i) with $\alpha \to 0$ (a vertical wall) we get $w = \dfrac{m(1+e)u}{M+m}$, the familiar answer for a direct collision with a free block of mass M.

- If $m \ll M$ the best angle has $\cos^2\alpha \to \frac{1}{2}$, i.e. $\alpha = 45°$, which is where $\sin\alpha\cos\alpha$ is greatest.
- Numerically, with $M = 3$, $m = 1$, $e = \frac{1}{2}$, $u = 4$, $\alpha = \frac{\pi}{6}$, part (ii) gives $w = \frac{2\sqrt{3}}{5} \approx 0.693$. A direct impulse calculation (solving for J from restitution) agrees with both formulas for random parameters, and a grid search over α confirms the maximising angle.

Common slips

- Forgetting that the cone moves, and writing the restitution equation as if the separation speed were just the particle's speed eu. The cone's velocity has a component $w\cos\alpha$ along the normal.
- Conserving momentum along the normal. The table's impulse has a component along $\mathbf{n}$, so only *horizontal* momentum is conserved.
- Taking the normal to make angle α with the vertical. It is the surface that makes angle α with the vertical, so the normal makes angle α with the horizontal.

Beyond STEP

The table here acts as a *constraint*: it removes one degree of freedom from the cone, and its impulse is whatever it has to be to enforce that. At university this becomes the theory of impulsive constraints, which is neatly handled with generalised coordinates in Lagrangian mechanics, and the projection onto the normal is the same idea behind the effective mass of a constrained body. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a careful treatment of collisions and impulses in two dimensions.

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